A Signature-Constrained Two-Stage Framework for Hyperspectral Target Detection Based on Generative Self-Supervised Learning

Haonan Qin, *Member, IEEE*, Shuai Wang, *Graduate Student Member, IEEE*, Yunsong Li, *Member, IEEE*, Weiying Xie, *Senior Member, IEEE*, Kai Jiang, and Kailang Cao

***Abstract*— Recently, the hyperspectral target detection (HTD) technique based on deep learning (DL) has been developed rapidly. However, existing algorithms show poor generalization across different hyperspectral images (HSIs), where repeated training and inference based on limited prior information are necessary. To liberate HTD from dependence on the quantity and the quality of training samples, this article proposes a signature-constrained two-stage framework for HTD (HTD-STF) based on generative self-supervised learning (GSSL). In the first stage for pretraining, to realize spectral–spatial reconstruction, we build an asymmetric autoencoder (AE) employing transformer blocks with long-range perception for generalized feature extraction. During pretraining, the spectral–spatial similarity loss (SSSL) is designed to improve the effect of reconstruction. In the second stage for fine-tuning and detection, the signature is utilized in preprocessing, training, and inference, respectively. Specifically, the coarse sample mining and tiling strategy in preprocessing not only facilitates the framework in flexible input dimension, but also provides pseudo-labels for end-to-end training. During training, we adopt the signature as guidance for feature-level fusion, which alleviates the impact of sample imbalance. After training, the final inference based on pixel-level fusion refines the original output. For ideal GSSL, the HyperMix-10K, a new large-scale hyperspectral dataset, has been constructed in this work, which contains numerous unlabeled HSIs captured in various scenes. Experimental results and analysis on real HSIs verify the effectiveness and generalization ability of the HTD-STF.**

***Index Terms*— Generative self-supervised learning (GSSL), hyperspectral image (HSI), hyperspectral target detection (HTD), remote sensing.**

I. Introduction

HYPERSPECTRAL imaging is a powerful technology that can simultaneously capture the spatial and spectral information of objects in a specific area [1]. The characteristic of high spectral resolution has enabled hyperspectral image (HSI) to effectively identify different kinds of materials through precise spectral analysis [2], which is a valuable tool for revealing essential characteristics of materials that are not visible to visual inspection [3]. Recently, as an important task of HSI analysis, hyperspectral target detection (HTD) has been extensively utilized in many fields such as military security [4], environmental monitoring [5], and resource exploration [6].

HTD generally refers to separating the desired target from the background and finding the number and location of target pixels in the HSI [7]. Specifically, HTD selects the target pixels from the background based on their inherent spectral differences, which usually uses the spectrum of the target as the prior information (i.e., signature) [8]. However, the target pixels usually occupy very small and sparse parts of the HSI. In addition, the background pixels usually include many complex and unknown types of ground material and some interfering substances different from ordinary noise [9]. Therefore, HTD is not only a simple binary classification problem but also a task that highlights the target component and simultaneously suppresses the background component [2].

In the past decades, a great number of HTD algorithms have been developed. Traditionally, HTD is achieved by simple spectral matching methods based on spectral similarity. The spectral angle mapper (SAM) [10] and spectral information divergence (SID) [11] measure the distance between the testing pixel and the target signature to get matching scores. Besides, structured background approaches, such as the orthogonal subspace projection (OSP) [12] model, assume that any mixed pixel can be regarded as a linear combination of several endmembers with different abundances.

Subsequently, researchers proposed many hypothesis testing methods based on statistical information and certain assumptions [13], which can be divided into two manners as follows.

1) *Under the Assumption of Distribution:* For example, the adaptive coherent estimator (ACE) [14] and the adaptive matched filter (MF) [15] are designed based on the unstructured background model. These detectors have been derived from a generalized likelihood ratio test.
2) *Under the Assumption of Minimization:* For instance, the constrained energy minimization (CEM) [16] and its variants, such as the E-CEM [17], have been developed to minimize the response of backgrounds and maximize the response of the targets in HSI.

Received 17 January 2025; revised 12 March 2025 and 3 April 2025; accepted 21 April 2025. Date of publication 24 April 2025; date of current version 2 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62121001, Grant 62322117, Grant 62371365, Grant 62471359, and Grant U24B20136; in part by the Fundamental Research Funds for the Central Universities under Grant XJSJ25001, Grant ZYTS24101, and Grant ZYTS24103; and in part by the Natural Science Basic Research Program of Shaanxi under Grant 2025JC-YBQN-879. *(Corresponding authors: Yunsong Li; Shuai Wang.)*

The authors are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: qinhaonan@xidian.edu.cn; 23011210728@stu.xidian.edu.cn; ysli@mail.xidian.edu.cn; wyxie@xidian.edu.cn; xdjiangkai@foxmail.com; caokailang@xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3564039

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

On this basis, some innovative techniques have been integrated into classical HTD methods. The kernel-based methods, such as the KOSP [18], employ kernel functions to map data into a high-dimensional space, facilitating the nonlinear transformation of the original features. Moreover, methods based on sparse and collaborative representations can obtain detection results by constructing an appropriate dictionary and calculating the residuals between reconstructed and original pixels. The typical examples include the combined sparse cooperative representation (CSCR) detector [19].

The above traditional methods always depend on assumptions that may not match the practical application scenarios. Recently, HTD models based on deep learning (DL) have gained attention and shown effectiveness due to the impressive representation ability of neural networks, such as autoencoder (AE), convolutional neural network (CNN) [20], generative adversarial network (GAN) [21], and transformer [22].

There are two main types of DL-based HTD models.

1) *Under Supervised Manner:* These methods mainly expand the target sample set via synthesis and develop an end-to-end detector by training on labeled pixel pairs [23]. For example, the HTD-Net [24] extends the CNN model through an improved AE and linear prediction strategy, using amounts of pseudo-labeled samples for training. The MLSN [25] leverages labeled source data and meta-learning techniques to learn how to discriminate between similarities and differences in spectra. The BLTSC [26] is trained on pseudo-labeled background samples to learn the background distribution. The HTD-TS^3 [27] and HTDFormer [28] based on the transformer, generate pseudo-labeled samples through a coarse detection strategy to better learn spectral–spatial features.
2) *Under Unsupervised Manner:* These methods train models without labeled samples but improve discrimination capabilities within specific constraints and then perform HTD with the simple matching strategy [23]. For example, the spectral regularized unsupervised network (SRUN) [29] is proposed to mine the potential information in HSI. The deep spatial–spectral network (DSN) [30] is designed to reduce the spatial and spectral redundancies simultaneously and provide strong discrimination. The interpretable representation network for HTD (HTD-IRN) [31] utilizes a lightweight multiscale transformer, which can achieve adaptive learning of the background subspace.

Although the above DL-based methods generally obtain better detection performance than traditional methods, there are some limitations existing. First, for a specific HTD task, the model is usually trained and tested on a fixed HSI captured in a fixed scene, making it scene-dependent. Second, repeated model training and parameter adjustment are required for different HSIs, which results in a highly time-consuming process. Besides, since HSIs often consist of thousands to millions of pixels, pixel-level manual annotation is time-consuming and labor-intensive. Most DL-based methods still need to be liberated from heavy dependence on the quantity and quality of the training samples.

Self-supervised learning (SSL) is a potential solution to address the above challenges. SSL is a kind of unsupervised learning that can mine self-supervised information from unlabeled data through large-scale pretraining [32]. Since massive available unlabeled HSIs are obtained daily by numerous in-orbit satellites [33], therefore, SSL can help model learn meaningful feature representations from unlabeled HSIs by utilizing the inherent attributes and relationships. According to different pretraining manners, SSL-based methods can be divided into contrastive SSL (CSSL) and generative SSL (GSSL) as follows.

1) *CSSL:* CSSL captures feature representations by minimizing the distance between positive samples and maximizing the distance between negative samples. Recently, some researchers have explored the application of CSSL in the HTD framework. The DPN-CSSTD [34] constructs a contrastive task using the HSI augmentation method and an encoder on unlabeled data, enabling the pretrained model with spectral–spatial discrimination ability. The SCLHTD [35] employs contrastive learning between the odd and even bands of unlabeled spectra to train a model that can distinguish spectral differences for HTD. The above methods mainly construct spectral-level contrast learning and obtain the ability to discriminate spectral similarity through feature extraction. However, they require carefully designed pretext tasks, which can be inefficient to prepare. For instance, data augmentation is necessary to generate and construct positive and negative sample pairs from unlabeled HSI for training.
2) *GSSL:* Unlike CSSL-based models, methods based on GSSL can extract meaningful features by recovering the original data distribution without assumptions. Recently, some researchers have explored the application of GSSL in remote sensing image processing. The SatMAE [36] is a pretraining model for multispectral images (MSIs) based on the masked AE (MAE) [37], which adopts a masked image modeling (MIM) framework and predicts the content of masked regions from visible regions. Although SatMAE can group adjacent spectral bands of multispectral data, it disrupts the continuity of the spectral dimension. The SpectralGPT [38] is pretrained on several multispectral datasets, which divides the input into 3-D cubes and creates a visual embedding for MAE. However, these models are not suitable for processing HSIs with hundreds of bands. Although Chen et al. [39] have proposed a pretraining model based on GSSL with spectral–spatial masking, it is pretrained on a dataset constructed for hyperspectral anomaly detection (HAD) and evaluated only on four HSIs. The HTD-WT [40] combines the wavelet transform with GSSL, but it is only pretrained on datasets generated by testing HSI.

Therefore, GSSL-based methods for HTD tasks are less explored. Inspired by GSSL, we intend to design a novel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

two-stage HTD framework that integrates GSSL and classical statistical detectors from the following two perspectives.

1) *Making Full Use of Limited Prior Information:* HTD aims to label each pixel as background or target in HSI with the supervision of preacquired target prior spectra [2]. The performance of HTD methods is susceptible to the prior target information, i.e., the theoretically pure and accurate spectrum of the target material [41]. In practice, obtaining adequate high-quality target prior spectra for HTD is labor-intensive or even impossible [2]. Therefore, further exploring the utilization of limited prior information in HSI is important and meaningful. In our opinion, the existing prior target signature needs to be fully used in the preprocessing, training, and inference stages. An optimal signature-based constraint may help the framework extract discriminative features during training and testing.
2) *Making Full Use of Data Without Prior:* The prior target signature can be obtained through three ways: acquiring prior spectrum from spectral libraries, acquiring prior spectrum from images, and collecting prior spectrum from actual scenes [42]. Due to the difficulty of acquiring more prior spectrum information, many DL-based methods focus on spectral feature extraction by expanding samples, resulting in low model transferability and high time consumption [34]. However, models only using prior signature information are insufficient to distinguish ground targets effectively [43]. Combined with the previous analysis, there is a strong need to design an HTD framework that can significantly reduce the demand for labeled or pseudo-labeled samples, effectively extract the spectral–spatial features, and can be flexibly transferred to different HTD tasks. Considering the frameworks based on GSSL are built by training on massive unlabeled datasets and optimizing numerous parameters, which leads to effective generalization across different scenarios, it is possible to develop a GSSL-based framework for HTD to solve the above problems.

To realize the above ideas, this article proposes a signature-constrained two-stage framework for HTD (HTD-STF) based on GSSL, which can liberate the HTD model from dependence on the quantity and quality of training samples and fully use the limited prior information.

As shown in Fig. 1, most existing HTD frameworks are trained separately on a specifically labeled HSI in the fixed scene, in which the feature extraction mechanism performs well only on training samples, but always shows poor generalization ability on samples from other HSIs with different scenes. Therefore, complete training in an end-to-end manner needs to be repeated sequentially when processing several HSIs. In contrast, the proposed HTD-STF exhibits stronger generalization capabilities. Benefiting from once pretraining with unlabeled HSIs, the pretrained encoder of HTD-STF can be adapted to different HSIs captured in various scenes in the fine-tuning stage. During the fine-tuning, HTD-STF utilizes a

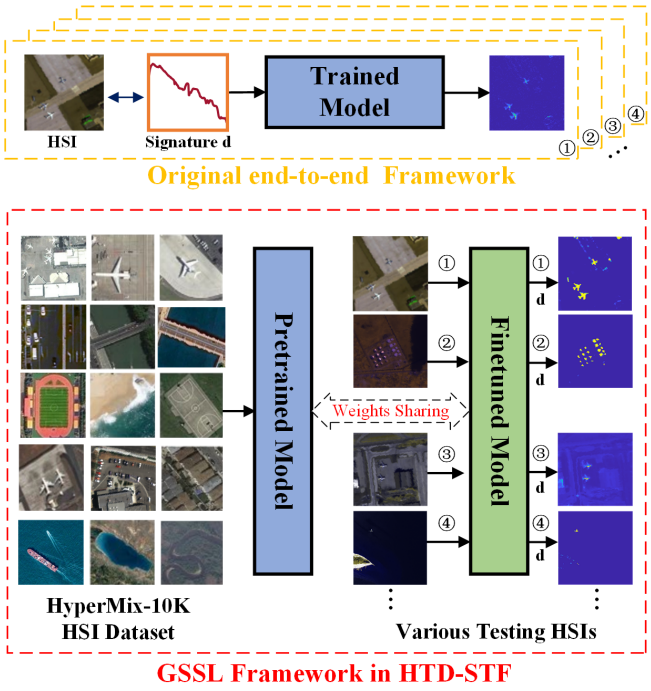


Fig. 1. Comparison of two different processing modes of HTD framework.

novel signature-constrained strategy for preprocessing, training, and inference to realize pixel-level detection finally.

The primary contributions can be summarized as follows.

1) To liberate DL-based methods from heavy dependence on the quantity and quality of the training samples, a novel GSSL-based two-stage HTD framework named HTD-STF, is proposed. Benefiting from GSSL, HTD-STF shows strong generalization when transferred to different HSIs captured in various scenes.
2) To provide GSSL with adequate unlabeled HSIs, a new large-scale hyperspectral dataset for GSSL, named HyperMix-10K, is constructed in this work, which integrates amounts of unlabeled HSIs, covering a vast range of natural and man-made scenes.
3) To explore the universal feature representation of diverse HSIs, an asymmetric AE architecture constrained by a novel spectral–spatial similarity loss (SSSL) is built in the first stage for pretraining, which is designed to learn key feature representations of HSIs.
4) To transfer the pretrained encoder to various HTD tasks, in the second stage for fine-tuning and detection, the target signature is innovatively utilized for preprocessing, training, and inference, which is adopted as the feature-level and pixel-level guidance to highlight targets.

The remainder of this article is structured as follows. Section II provides a detailed description of HTD-STF. Section III mainly describes the experimental results and corresponding analysis. Finally, the conclusion is drawn in Section IV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

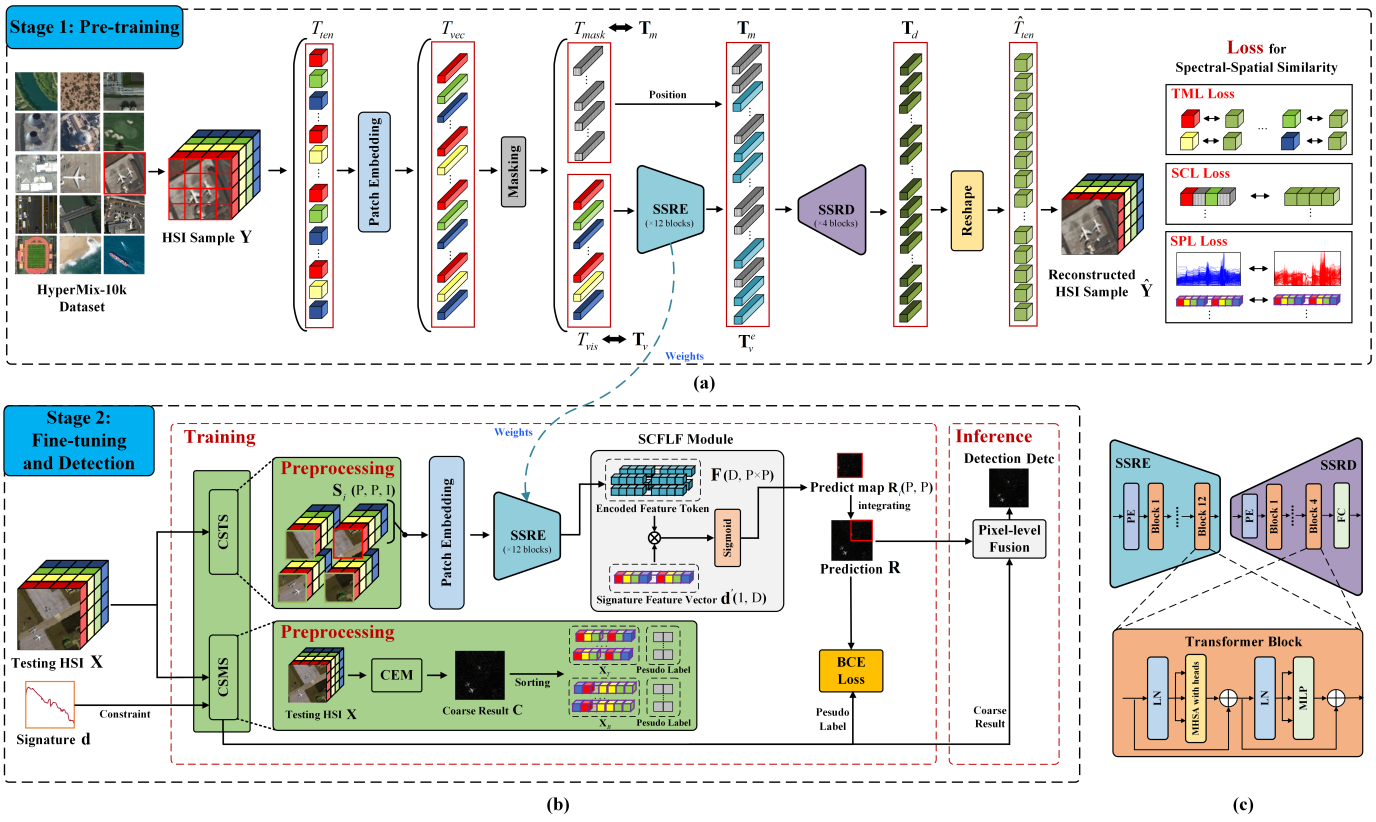


Fig. 2. Overall architecture of the proposed HTD-STF framework. (a) In the pretraining stage based on GSSL, the input HSI sample $\mathbf{Y}$ is first modeled as a series of 3-D spectral–spatial tensors, denoted as T_{ten}. And the patch embedding operation is adopted to transfer each tensor into D dimension, producing a series of embedded vectors, denoted as T_{vec}. Then, T_{vec} will be randomly masked to obtain two groups of embedded vectors, denoted as T_{vis} and T_{mask}, respectively, which will be flattened and concatenated to produce the visible and masked embedded token, denoted as $\mathbf{T}_v$ and $\mathbf{T}_m$, respectively. Next, only the visible embedded token $\mathbf{T}_v$ is used as the input of the SSRE, which produces the output as the encoded feature token $\mathbf{T}_v^e$. Finally, the encoded feature token $\mathbf{T}_v^e$ and the masked embedded feature token $\mathbf{T}_m$ are fed to the SSRD to produce the decoder output $\mathbf{T}_d$, which can be reshaped to obtain the reconstructed 3-D spectral–spatial tensors $\hat{T}_{\text{ten}}$. (b) In the fine-tuning and detection stage for the testing HSI, the input testing HSI $\mathbf{X}$ is preprocessed with the signature-constrained sample mining and tiling strategy, which contains the CSMS and the CSTS. First, the CSMS performs the coarse detection on the testing HSI $\mathbf{X}$ to obtain the coarse detection result $\mathbf{C}$ and the pseudo-label sample set, including the pseudo-target sample set $\mathbf{X}_T$ and the pseudo-background sample set $\mathbf{X}_B$. And the CSTS transfers the testing HSI $\mathbf{X}$ into a HSI patch sequence S. During the training, each HSI patch $\mathbf{S}_i$ is processed with the patch embedding operation and fed to the pretrained SSRE, which can produce the encoded feature token $\mathbf{F}$. Then, the prior target feature vector $\mathbf{d}'$ and the encoded feature token $\mathbf{F}$ are fed into the SCFLF module to obtain the predict map $\mathbf{R}_i$. And each predict map $\mathbf{R}_i$ is integrated back to produce the prediction result $\mathbf{R}$. Finally, during inference, the signature-constrained pixel-level fusion combines the coarse detection result $\mathbf{R}$ and the prediction result $\mathbf{R}$ via an exponential-constrained nonlinear function to produce the detection result **Detc**. (c) SSRE module and SSRD module make up the asymmetric AE architecture for spectral–spatial reconstruction, which consists of the stacked transformer blocks.

II. Proposed Two-Stage HTD Framework

This section demonstrates the proposed HTD framework, which consists of two processing stages: 1) the pretraining stage based on GSSL and 2) the fine-tuning and detection stage for the testing HSI. Fig. 2 shows the details of the two stages. Different from traditional HTD frameworks, HTD-STF utilizes once large-scale pretraining on the HyperMix-10K dataset, which can enable the model to extract universal feature representation based on the inherent attributes and relationships of unlabeled HSIs. After a single pretraining, HTD-STF can be transferred to different testing HSIs through the simple and fast fine-tuning based on the signature-constrained strategy.

A. Pretraining Stage Based on GSSL

In this stage, an asymmetric AE architecture is designed to realize spectral–spatial reconstruction, which employs stacked transformer blocks as the backbone. Following the principle of MIM from the MAE [37], the input HSIs are modeled as visible or masked 3-D tensors through the spectral–spatial masking strategy. Then, the encoder for spectral–spatial reconstruction (SSRE) mapped these visible tensors to encoded feature vectors. And the decoder for spectral–spatial reconstruction (SSRD) reconstructs the original tensors based on these encoded feature vectors and these masked tensors. By performing effective GSSL on the HyperMix-10K dataset constructed in this article, the feature extraction ability of the proposed framework is enhanced by optimizing the proposed SSSL.

Specifically, this stage consists of three components: spectral–spatial masking, AE for spectral–spatial reconstruction, and a loss function based on spectral–spatial similarity.

1) Spectral–Spatial Masking: Traditional HTD algorithms typically process a single HSI as an independent HTD task, in which the pixels are regarded as the training samples. Differently, the pretraining in HTD-STF takes a large number of HSIs as image-level samples for training. To utilize various

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

HSIs as the input of HTD-STF, a spectral–spatial masking strategy is designed at first.

HSIs often consist of hundreds of bands that provide rich spectral information. However, the dimensionality disaster will make the pretraining on large-scale HSIs challenging. Therefore, the dimensionality reduction is performed in the proposed masking strategy. Let an C-channel HSI be denoted as $\mathbf{Y}' \in \mathbb{R}^{H\times W\times C} = \{\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_C\}$ where H and W represent the number of rows and columns in spatial dimension, and C is the number of spectral channels. To ensure different HSIs have the same number of channels during the pretraining, L continuous channels of $\mathbf{Y}'$ are selected to acquire $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_l, \ldots, \mathbf{Y}_L]$, where $1 \leqslant l \leqslant L$, $L < C$ and L is fixed for the pretraining dataset.

On this basis, each HSI will be modeled as a series of 3-D spectral–spatial tensors without overlapping, which ensures the continuity of spectral–spatial information. And the patch embedding operation is adopted to transfer each 3-D tensor into a specific embedding dimension D. Specifically, given an HSI sample $\mathbf{Y} \in \mathbb{R}^{H\times W\times L}$ from the pretraining dataset consisting of HSIs with a fixed number (such as L) of channels, the 3-D convolution is applied to process the input HSI $\mathbf{Y}$ along its spatial and spectral dimensions. Note that the kernel size and stride of 3-D convolution are both set as (s, s, d). Then, a sequence of 3-D spectral–spatial tensors can be obtained as $T_{\text{ten}} = \{\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_q, \ldots, \mathbf{T}_{(H/s)\times(W/s)\times(L/d)}\}$. Each tensor $\mathbf{T}_q \in \mathbb{R}^{s\times s\times d}$ in T_{ten} will be projected into a D-dimensional embedded vector in the embedding space, which will produce a sequence of embedded vectors, denoted as T_{vec}.

Furthermore, the embedded vectors in T_{vec} are randomly masked, which can obtain two groups of embedded vectors, denoted as T_{vis} and T_{mask}, respectively. Only the visible embedded vectors will be fed into the SSRE for training. In practice, the groups of embedded vectors are flattened and concatenated, which can produce the visible and the masked embedded feature token, denoted as $\mathbf{T}_v$ and $\mathbf{T}_m$, respectively.

The above procedures can be summarized as formulas

$$T_{\text{vec}} = \text{PatchEmbeding}(T_{\text{ten}}) \tag{1}$$

$$\{T_{\text{vis}}, T_{\text{mask}}\} = \text{mask}(T_{\text{vec}}) \tag{2}$$

$$\mathbf{T}_v = \text{Concat}(T_{\text{vis}}), \mathbf{T}_v \in \mathbb{R}^{G\times D} \tag{3}$$

$$\mathbf{T}_m = \text{Concat}(T_{\text{mask}}), \mathbf{T}_m \in \mathbb{R}^{B\times D} \tag{4}$$

where G and B represent the number of visible and masked vectors in the embedded vectors, respectively.

As shown in Fig. 2(a), by introducing masking randomness, each HSI can be randomly transformed into visible and masked embedded features in each iteration, allowing the model to adapt to input data variations in different scenarios. Note that the masking rate is a critical parameter that could affect detection performance. And comprehensive experiments with analysis in Section III are employed to discuss and identify the optimal masking rate.

2) AE for Spectral–Spatial Reconstruction: To realize GSSL based on unlabeled HSIs, an AE architecture is designed for spectral–spatial reconstruction, which consists of SSRE and SSRD. Note that the input of SSRE and SSRD is different. Inspired by the vision transformer [44], the proposed AE architecture employs a stacked transformer as the basic block. Fig. 2(c) shows the details of the SSRE module and SSRD module based on transformer blocks.

1) *SSRE*: SSRE consists of 12 transformer blocks with 12 attention heads. The visible embedded feature token $\mathbf{T}_v$ is used as the input of the SRRE, which will be mapped to the encoder dimension D. First, the position embedding (PE) operation is applied to $\mathbf{T}_v$ to retain position information of each embedded vector in $\mathbf{T}_v$, which can be formulated as follows:

$$\mathbf{T}_v^p = \mathbf{T}_v + \mathbf{T}_{vp} = \mathbf{T}_v + \text{PE}(\mathbf{T}_v) \tag{5}$$

where $\mathbf{T}_{vp} \in \mathbb{R}^{G\times D}$ represents the learned positional matrix.

After PE, the visible embedded feature token with position information, denoted as $\mathbf{T}_v^p$, is fed into the SSRE module for feature extraction based on the self-attention (SA) mechanism, which is the same as HTD-TS^3 [27].

As shown in Fig. 2(c), each transformer block in SSRE consists of three operations: layer normalization (LN), multihead SA (MHSA), and multilayer perceptron (MLP).

Specifically, the LN layer not only reduces computing time, but also alleviates the issues of disappearance or explosion. The MHSA captures the relations of the input sequences. The MLP is used for feature to realize a linear transformation between two SA layers. The calculation for the SA of each head in MHSA can be formulated as follows:

$$\text{SA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_t}}\right)\mathbf{V} \tag{6}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the query matrix, key matrix, and value matrix, respectively, and d_t is the dimension of the input data. The MHSA with J heads can be formulated as follows:

$$\text{MHSA}\big(\text{LN}\big(\mathbf{T}_v^p\big)\big) = \text{Concat}\big(\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_j, \ldots, \mathbf{h}_J\big)\mathbf{W}^o \tag{7}$$

where $\mathbf{h}_j = \text{SA}(\text{LN}(\mathbf{T}_v^p)\mathbf{W}_j^Q, \text{LN}(\mathbf{T}_v^p)\mathbf{W}_j^K, \text{LN}(\mathbf{T}_v^p)\mathbf{W}_j^V)$, and $\mathbf{W}$ represents the corresponding learned parameter matrix.

Therefore, the output of SSRE can be obtained as the encoded feature token $\mathbf{T}_v^e$, which can be expressed as follows:

$$\mathbf{T}_v^e = \text{SSRE}\big(\mathbf{T}_v^p\big), \mathbf{T}_v^e \in \mathbb{R}^{G\times D}. \tag{8}$$

2) *SSRD*: Rather than the symmetric structure of classic AE architecture, the asymmetric structure of AE often produces better results in GSSL [37]. Therefore, different from SSRE, SSRD is designed as shown in Fig. 2(c), which consists of four transformer blocks with 16 attention heads.

As shown in Fig. 2(a), the proposed SSRD reconstructs the original HSI using the encoded feature token $\mathbf{T}_v^e$ from SSRE and the masked embedded feature token $\mathbf{T}_m$ without encoding. Note that the embedded vectors in $\mathbf{T}_m$ need to be added into $\mathbf{T}_v^e$ based on their original positions.

SSRD first maps the input vectors to a decoder dimension D' and feeds these vectors into the transformer blocks. As for the final layer of the SSRD, a fully connected (FC) layer is employed as the prediction head to project each decoded vector into a specific dimension, which can be reshaped to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

the original 3-D spectral–spatial tensor. Finally, a reshaping operation is performed on $\mathbf{T}_d$, i.e., the output of SSRD, to obtain the sequence of reconstructed 3-D spectral–spatial tensors, denoted as $\hat{T}_{\text{ten}}$. The process of SSRD can be expressed as follows:

$$\mathbf{T}_d = \text{SSRD}\left(\mathbf{T}_v^e, \mathbf{T}_m\right), \mathbf{T}_d \in \mathbb{R}^{(G+B)\times(s\times s\times d)} \tag{9}$$

$$\hat{T}_{\text{ten}} = \text{Reshape}(\mathbf{T}_d). \tag{10}$$

3) Spectral–Spatial Similarity Loss: To realize the ideal reconstruction of input HSI samples during the pretraining, the objective function needs to be designed to accurately measure the loss value of spectral–spatial similarity between each original HSI and its corresponding reconstructed version. We define the SSSL in this article, which consists of three components: the tensor matching loss (TML), the spectral continuity loss (SCL), and the spectral perception loss (SPL).

First, for each HSI, the TML is employed to evaluate the quality of tensor-level reconstruction between the original 3-D spectral–spatial tensors T and reconstructed ones $\hat{T}$, which can be calculated by the mean square error (mse). The TML function can be calculated as follows:

$$\mathcal{L}_{\text{TML}} = \frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{T}_n - \hat{\mathbf{T}}_n\right)^2 \tag{11}$$

where $N = (H/s) \times (W/s) \times (L/d) = G + B$ represents the number of 3-D spectral–spatial tensors in an input tensor sequence $\{\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_n, \ldots, \mathbf{T}_N\}$.

Second, the SCL is employed to evaluate the spectrum-level reconstruction quality, which can be calculated as the reconstruction error of the spectral content between the original and reconstructed spectra in a tensor group, i.e., T'_m and $\hat{T}'_m$. The SCL function can be calculated as follows:

$$\begin{aligned}\mathcal{L}_{\text{SCL}} &= \frac{1}{M}\sum_{m=1}^{M}\left(T'_m - \hat{T}'_m\right)^2 \\ &= \frac{1}{M}\sum_{m=1}^{M}\left(\left[\mathbf{T}_1, \ldots, \mathbf{T}_{\frac{L}{d}}\right]_m - \left[\hat{\mathbf{T}}_1, \ldots, \hat{\mathbf{T}}_{\frac{L}{d}}\right]_m\right)^2\end{aligned} \tag{12}$$

where $M = (H/s) \times (W/s)$ represents the number of tensor groups. Note that each tensor group consists of (L/d) 3-D spectral–spatial tensors along with the spectral dimension of the input HSI.

Third, to further quantify the quality of spectral–spatial reconstruction, the SPL is proposed to perceive the difference between the original and the reconstructed spectra, which can be calculated by comparing the vector variance of the original and reconstructed spectra in a tensor group. The SPL function can be calculated as follows:

$$\mathcal{L}_{\text{SPL}} = \frac{1}{M}\sum_{m=1}^{M}\left(V_m^{\sigma} - \hat{V}_m^{\sigma}\right)^2 \tag{13}$$

where the V_m^{σ} and $\hat{V}_m^{\sigma}$ represent the vector variance of the original and the reconstructed spectra.

The SPL function can handle the entire spectral range of each tensor group, allowing for comparison based on the vector variance difference between the reconstructed and original spectra. By penalizing sharp variations among adjacent bands, SPL can effectively suppress the abrupt changes that may occur during the spectra reconstruction.

Therefore, the final objective function can be described as follows:

$$\mathcal{L}_{\text{SSSL}} = \alpha\mathcal{L}_{\text{TML}} + \beta\mathcal{L}_{\text{SCL}} + \lambda\mathcal{L}_{\text{SPL}} \tag{14}$$

where α, β, and λ are used to balance the contributions of the corresponding terms in the objective function.

B. Fine-Tuning and Detection Stage

After the first stage, the pretrained SSRE can extract a universal feature representation of HSIs. During the fine-tuning and detection stage, the pretrained SSRE is transferred to a specific HTD task through the simple and fast fine-tuning based on three signature-constrained strategies: the sample mining and tiling, the fine-tuning, and the inference fusion.

1) Signature-Constrained Sample Mining and Tiling: Considering the sparsity of target pixels and the abundance of background pixels in HSIs, the category imbalance in HSI can greatly affect the detection performance. In addition, since the size of the testing HSI is not fixed, the transformer block often fails to work due to the memory overflow when the input size is too large. To address these issues, a signature-constrained sample mining and tiling strategy is proposed, which consists of two parts: the coarse sample mining strategy (CSMS) and the coarse sample tiling strategy (CSTS).

1) *CSMS*: Inspired by HTDFormer [28], the classic detector CEM [16] is applied to perform coarse detection on the testing HSI $\mathbf{X}$, which can produce the coarse result $\mathbf{C}$ with pseudo-labels for subsequent training and inference. Specifically, we obtain the target signature $\mathbf{d}$ from spectral libraries, constrained by $\mathbf{d}$, the CEM exploits the spectral differences between the targets and background pixels via a finite impulse response (FIR) linear filter $\mathbf{w}$, which keeps the output of $\mathbf{d}$ as a constant, i.e., $\mathbf{d}^{\mathbf{T}}\mathbf{w} = 1$. The average filter output energy of the data, i.e., $E = \mathbf{w}^{\mathbf{T}}\mathbf{R}\mathbf{w}$, is suppressed, where $\mathbf{R}$ is the covariance matrix of $\mathbf{X}$. The optimal solution can be derived

$$\mathbf{w}^{\text{opt}} = \frac{\mathbf{R}^{-1}\mathbf{d}}{\mathbf{d}^{\mathbf{T}}\mathbf{R}^{-1}\mathbf{d}}. \tag{15}$$

The mining of training samples can be realized by constructing the pseudo-background sample set $\mathbf{X}_B$ and the pseudo-target sample set $\mathbf{X}_T$. The detailed steps are as follows.

1) According to the values in the coarse result $\mathbf{C}$, all pixels from the testing HSI $\mathbf{X}$ are sorted from low to high, which can be regarded as candidate samples.
2) Then, the top 50% pixels and the last 1.5% pixels are selected to form the pseudo-background sample set $\mathbf{X}_B$ and the pseudo-target sample set $\mathbf{X}_T$, respectively.

After utilizing the CSMS, the potential pixels can be filtered out and assigned with pseudo-labels. However, note that the pseudo-sample sets constructed based on coarse detection from limited prior target spectral information may be incompletely and inaccurately labeled. Therefore, the fine-tuning and detection stage can also be considered as a process under inaccurate supervision, one of the three types of weak supervision [45]. Besides, the selection ratio of pseudo-background pixels is a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

critical parameter that could affect the detection performance. Comprehensive experiments with analysis in Section III are employed to identify the optimal selection ratio.

2) *CSTS*: To facilitate the pretrained SSRE in processing input dimensions flexibly, the original testing HSI is segmented into smaller HSI patches, which can avoid transformer work failure due to the memory overflow and improve the generalization of the model.

Specifically, given a testing HSI $\mathbf{X}$ with I channels, it is split into overlapping square HSI patches with a side size of P pixels. And the overlapping distance is set as $P/2$ pixels. Note that the size P is selected as a hyperparameter. Therefore, a sequence S consisting of HSI patches can be obtained.

The above processing can be described as follows:

$$S = \text{Slice}(\mathbf{X}) \tag{16}$$

where $S = [\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_i, \ldots]$ and $\mathbf{S}_i \in \mathbb{R}^{P\times P\times I}$.

2) Signature-Constrained Fine-Tuning: For the testing HSI $\mathbf{X}$, each HSI patch $\mathbf{S}_i$ of sequence S will be processed and trained under the following procedures in this stage.

Given an HSI patch $\mathbf{S}_i$, it will be transformed into $P \times P$ embedded vectors at first through the patch embedding operation. These embedded vectors are then flattened and fed into the pretrained SSRE to produce the feature token $\mathbf{F}$ consisting of $P \times P$ encoded feature vectors of $(D, 1)$ dimension. Note that the pretrained SSRE serves as a feature extractor, which can be directly used for fine-tuning, as the pretrained SSRE processes embedded vectors sequentially rather than all at once, and each embedded vector has the same dimension.

After SSRE, the MLP layer is usually adopted to predict whether the label of each encoded feature vector belongs to the target category. However, we innovatively propose a signature-constrained feature-level fusion (SCFLF) module, which can highlight the target information based on the prior target signature. Specifically, the prior target signature $\mathbf{d}$ is projected into the D-dimensional embedding space through the linear mapping, which produces a signature feature vector $\mathbf{d}'$ of $(1, D)$. Then, this signature feature vector is directly multiplied with each encoded feature vector in the feature token to acquire the primary predict map, which consists of $P \times P$ confidences (to be target) corresponding to $P \times P$ pixels. And a sigmoid function is subsequently applied, which can constrain each probability of the corresponding pixel being target to the range of $(0, 1)$, yielding the predict map $\mathbf{R}_i$ of the input HSI patch. The above process can be described as follows:

$$\mathbf{R}_i = \mathbf{d}' * \mathbf{F} \tag{17}$$

where $*$ represents the inner product between $\mathbf{d}'$ and each encoded feature vector in $\mathbf{F}$. Note that the product reflects the similarity between the two vectors, with higher values indicating greater similarity.

The SCFLF helps model identify as many target pixels as possible, which can implicitly prevent the gradients from being dominated by the background pixels, effectively mitigating bias introduced by sample imbalance.

When all HSI patches are processed according to the above procedures, each predict map $\mathbf{R}_i$ will be integrated back to produce the prediction result $\mathbf{R}$ with the same dimension as the testing HSI, which can be conducted based on the original position information of each HSI patch.

During the fine-tuning, the parameters of SSRE are inherited from the pretraining stage and are trainable in the fine-tuning stage, which can retain the robust spectral–spatial feature extraction ability. And the binary cross-entropy (BCE) is adopted to realize the calculation of loss value for pixel-level binary classification. Note that the pseudo-label samples participating in the computation are from the CSMS. The object function can be expressed as follows:

$$\mathcal{L}_{\text{BCE}} = \frac{1}{\text{R}} \sum_{\text{r}=1}^{\text{R}} -\left[y_{\text{r}} \cdot \log\left(p_{\text{r}}^1\right) + (1 - y_{\text{r}}) \cdot \log\left(p_{\text{r}}^2\right)\right] \tag{18}$$

where p_r^1 and p_r^2 are the prediction possibility of sample $\mathbf{x}_r$ to be target and background. And y_r represents the pseudo-label of sample $\mathbf{x}_r$ with values 1 for target or 0 for background. Besides, R is the number of pseudo-samples.

3) Signature-Constrained Inference Fusion: Based on the fine-tuning, to further highlight the target information, a signature-constrained detector is developed to construct the pixel-level inference fusion under background suppression (BS).

Specifically, the coarse detection result C with spatial distribution information is hierarchically fused with the prediction result $\mathbf{R}$ via an exponential constrained nonlinear function. The inference based on the pixel-level fusion enables the integration of features from the spectral domain and spatial domain while suppressing background interference.

In this way, the detection result **Detc** can be acquired as follows:

$$\mathbf{Detc} = \kappa \odot \mathbf{R} = \left(\mathbf{1} - e^{-0.05\cdot\theta\cdot\mathbf{C}}\right) \odot \mathbf{R} \tag{19}$$

where θ is a positive parameter used to adjust the nonlinear transformation factor κ, which aims to use the coarse detection result $\mathbf{C}$ to suppress the background.

After the inference fusion, pixels with high probability of being target are retained, and κ is relatively stable and close to 1. However, pixels with a high likelihood of being background are suppressed, and κ drops sharply and approaches 0. Fig. 3 shows the shape of κ under different θ selections.

III. Experiments

A. Datasets Description

1) Pretraining Dataset: Existing large-scale remote sensing datasets are predominantly focused on high-resolution RGB imagery with limited spectral information, which makes it challenging to extract discriminative details of targets.

In the pretraining stage, to realize GSSL for spectral–spatial feature extraction of hyperspectral data, a new large-scale hyperspectral dataset, HyperMix-10K, is proposed in this work. HyperMix-10K consists of 10, 247 HSI samples covering numerous real scenes, more than the popular ABU dataset [46], which only includes 13 testing scenes. All samples in the HyperMix-10K dataset are captured by the Airborne

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

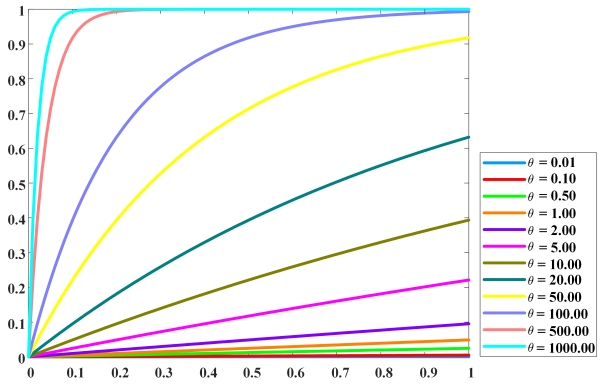


Fig. 3. Shape of exponential constrained nonlinear transform factor κ.

Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, and cropped into a uniform size of 100 × 100 with 224 spectral bands covering wavelengths ranging from 0.4 to 2.5 μm.

Fig. 4 demonstrates the diversity of targets in HyperMix-10K, which contains abundant categories including airplanes, vehicles, boats, bridges, buildings, church, stadium, palace, snow mountain, football field, basketball court, swimming pool, storage tank, oil well, pipeline, island, intersection, park, desert chaparral, thermal power station, and so on.

As for the background, HSIs from HyperMix-10K are captured in various scenes, as shown in Fig. 5, including airport, grassland, forest, farmland, desert, lake, river, harbor, coast, parking lot, industrial area, commercial area, freeway, terrace, wetland, and so on. Fig. 6 shows the distribution of background categories in the dataset. It can be observed that the largest category is the commercial area, followed by the industrial area and the mountain. And smaller categories mainly include bridge, harbor, and wetland. This diversity ensures that HyperMix-10K covers a wide range of real-world environments.

2) Fine-Tuning Datasets: During the fine-tuning, eight real HSIs for HTD tasks are used to evaluate the performance of the proposed HTD-STF, including five HSIs from the ABU datasets as well as three HSIs from the San Diego Dataset, the Hyperspectral Digital Imagery Collection Experiment (HYDICE) Dataset, and the EI Segundo Dataset, respectively.

ABU Datasets: These HSI datasets [46] were acquired by the AVIRIS sensor, which consists of the airport scene, the beach scene, and the urban scene. Five HSI datasets of ABU datasets are selected for the following experiments, which can be denoted as: Airport-1 dataset, Airport-2 dataset, Beach dataset, Urban-1 dataset, and Urban-2 dataset, respectively. After removing the water-absorption and noisy bands, the sizes of each HSI dataset are 100 × 100 × 205, 100 × 100 × 191, 150 × 150 × 188, 100 × 100 × 204, and 100 × 100 × 207, respectively. And the target of each HSI dataset occupies 87, 60, 19, 67, and 155 pixels, respectively.

San Diego Dataset: This HSI dataset [17] was acquired by the AVIRIS sensor, covering the San Diego Airport region in CA, USA. The dataset consists of 100 × 100 pixels, with a spatial resolution of 3.5 m. And it includes 189 spectral bands with wavelength ranging from 370 to 2510 nm. The scene depicts an airport with the targets of three airplanes occupying 134 pixels.

HYDICE Dataset: This HSI dataset [46] was captured by the HYDICE airborne sensor from the HYDICE over urban areas in CA, USA. The urban scene consists of 80 × 100 pixels, each with 175 spectral channels and a wavelength range from 400 to 2500 nm. The spatial resolution of the image is 1 m. The scenes mainly includes a vegetation zone, a building zone, and several roads. Man-made objects, such as cars and rooftops, occupying 19 pixels, are selected as targets.

EI Segundo Dataset: This HSI dataset [47] was captured by the AVIRIS sensor, which has a size of 250 × 300 pixels, 224 spectral bands, a spatial resolution of 7.1 m, and a wavelength range from 366 to 2496 nm. The images were captured in a region of EI Segundo, CA, USA, with scenes including an area of a refinery, several residential areas, and parks. This dataset selects storage tanks and towers within the refinery as targets, which occupy 2048 pixels.

Pavia Dataset: This HSI dataset [48] was captured by the reflective optics system imaging spectrometer (ROSIS) sensor, which has a size of 150 × 150 pixels, 102 spectral bands, a spatial resolution of 1.3 m, and a wavelength range from 430 to 960 nm. The images were captured in a region of Lombardia, Italy, with scenes including an area of beach and water, several vehicles. This dataset selects some vehicles as targets, which occupy 68 pixels.

B. Experimental Settings

1) Implementation Details: For the proposed HTD-STF framework, in the pretraining stage, the input channels L after the dimensionality reduction is 210. The masking rate is set to 0.7 and the learning rate is set to $2e^{-4}$. The training epoch is set to 200. Encoder and decoder dimension D and D' are set to 768 and 512, respectively. During the fine-tuning and detection stage, the learning rate is set as $6e^{-5}$. And the training epoch is set as 30. Besides, the selection ratio of pseudo-background pixels is set as 0.5. And the HSI patch size P is set as 50. Note that the optimizer AdamW is used during the two stages, and the parameters α, λ, β, and θ are set as 0.1, 0.001, 100, and 0.0001, respectively. All the experiments were implemented on an Ubuntu 22.10 LTS operating system (OS) with hardware configuration of a six-core 12th Gen Intel[1] Core[2] i5-12400F CPU operating at 2.5 GHz and 32-GB RAM and two NVIDIA GeForce RTX 3090 GPUs with 24-GB memory. Compiling softwares consist of a Visual Studio Code 2024 version 1.95 and a MATLAB R2023a.

2) Comparative Methods: To evaluate the performance of the proposed HTD-STF framework, eight target detection methods are selected for comparison. Especially, there are five traditional methods including MF [15], SAM [10], CEM [16], E-CEM [17], SID [11], and four DL-based methods containing BLTSC [26], HTD-IRN [31], MLSN [25], and HTD-WT [40]. These methods are popular as their outstanding performance. And there are parameter settings for the competitive methods, excluding MF, SAM, and SID, as these methods have

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

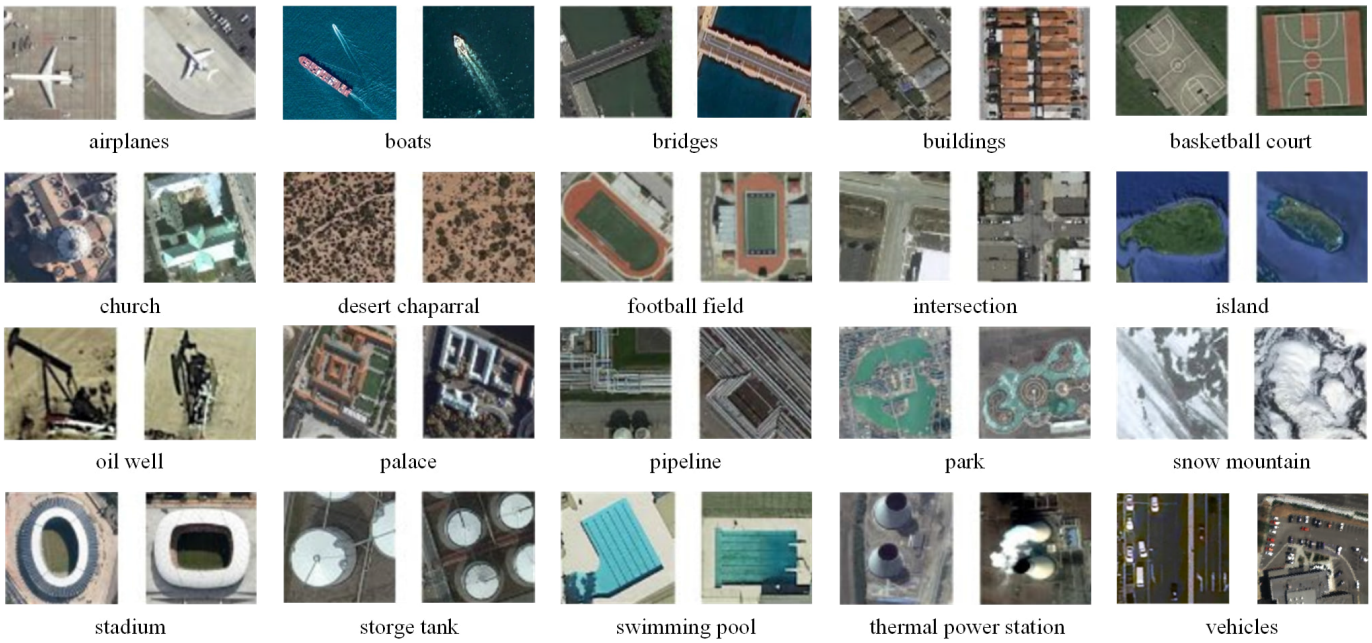


Fig. 4. Visualization examples of abundant categories of target in the HypermMix-10K dataset.

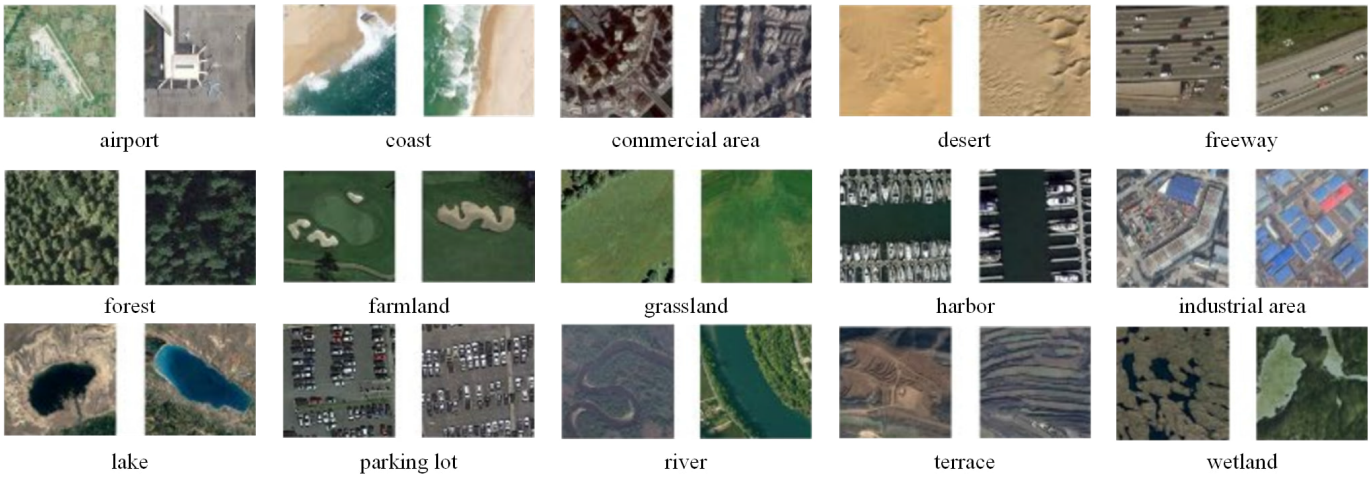


Fig. 5. Visualization examples of various scenes of background in the HypermMix-10K dataset.

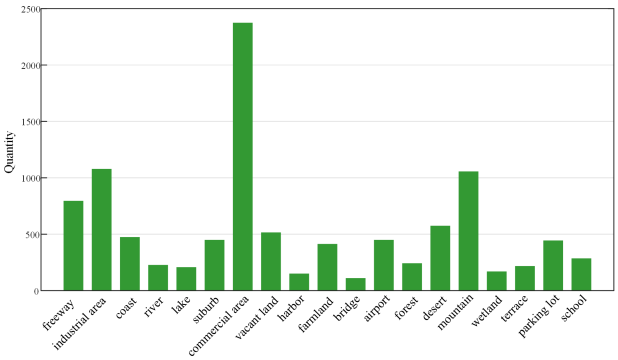


Fig. 6. Distribution of background categories in HyperMix-10K dataset.

no adjustable parameters. Then, for E-CEM, the number of detection layers, the number of CEM detectors per layer, and the regularization coefficient are set to 10, 6, and $1e^{-6}$, respectively. For BLTSC, the learning rate and epoch are set to $1e^{-4}$ and 500. For HTD-IRN, the learning rate and epoch are set to $1e^{-3}$ and 5000. For MLSN, the learning rate and epoch are set to $1e^{-3}$ and 50. For HTD-WT, the parameter λ and epoch are set to $1e^{-2}$ and 100.

3) Evaluation Metrics: Several metrics are utilized to evaluate the performance of the proposed framework: the receiver operating characteristic (ROC) curve [49], the area under the ROC curve (AUC) [50], the target-background separability box plots [47], as well as the training and testing time of the model.

Specifically, the ROC curve has been widely used as an evaluation tool for the HTD task, which describes the varying relationship between the true positive rate (TPR) and false positive rate (FPR) based on the reference map. The ROC curve can be estimated via plotting the TPR and FPR at different thresholds denoted as τ. The AUC metric is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

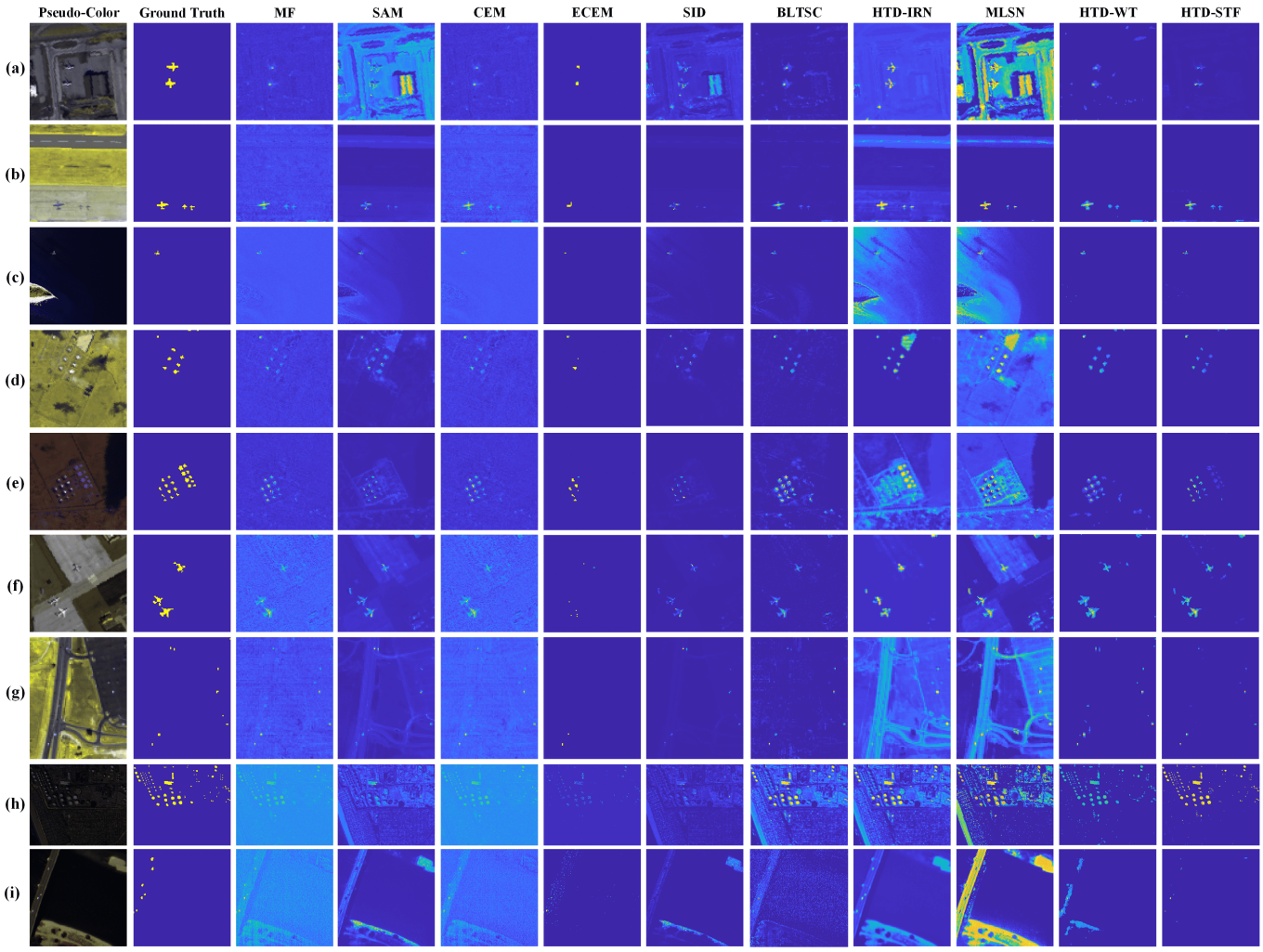


Fig. 7. Visualization of detection results of ten detection methods on nine hyperspectral datasets for HTD task. (a) Airport-1. (b) Airport-2. (c) Beach. (d) Urban-1. (e) Urban-2. (f) San Diego. (g) HYDICE. (h) EI Segundo. (i) Pavia.

considered as a complete description of HTD detection performance. Specifically, $\mathrm{AUC}_{(D,F)}$ and $\mathrm{AUC}_{(F,\tau)}$ are important metrics for assessing detector performance, which reflects the effectiveness, the detection probability, and the BS capability of a detector, respectively. Generally, the higher $\mathrm{AUC}_{(D,F)}$ and the smaller $\mathrm{AUC}_{(F,\tau)}$ indicate better detector performance. Note that we use AUC_1 and AUC_2 to represent the $\mathrm{AUC}_{(D,F)}$ value and the $\mathrm{AUC}_{(F,\tau)}$ value, respectively.

To further evaluate the separation of the target and background, the target-background separability plot [47] is adopted, in which the corresponding values of the target and background are extracted from the detection results. The red box represents the target, and the green box represents the background. The larger the distance between the two colored boxes, the higher the separation of the target and background.

C. Detection Performance

To evaluate the performance of the proposed HTD-STF framework, eight detection methods are used for comparison, including six traditional methods and two DL-based methods. The effectiveness of the proposed HTD-STF framework is verified by detection accuracy and computing speed.

Fig. 7 visualizes the detection results obtained by the proposed HTD-STF and eight other competing methods on eight real HSI datasets. Specifically, the SID and the E-CEM achieve BS relatively well at the cost of incomplete target detection. Traditional methods such as the CEM, the SAM, and the MF can detect most of the target pixels, but the CEM and the MF are less effective in suppressing background interference than the SAM model. As for the four DL-based methods, the HTD-IRN and MLSN can detect more target pixels of interest. However, the BLTSC and HTD-WT can suppress more background interference.

Compared with other methods, the proposed HTD-STF achieves a satisfactory balance between highlighting the target and suppressing the background. For instance, on the Urban-2 dataset, the HTD-STF can detect more interesting target pixels in the right zone while suppressing other background pixels. Besides, on the EI Segundo dataset, target pixels are relatively scattered but cover a significant part of the scene. As shown in Fig. 6, the HTD-STF achieves a better separation between targets and background. In contrast, the other methods on the EI Segundo dataset show poor BS performance with severe missed detections. Therefore, the HTD-STF can focus

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
AUC VALUES OF TEN DETECTION METHODS ON FOUR HYPERSPECTRAL DATASETS FOR HTD TASK

Methods	Airport-1		Airport-2		Beach		Urban-1	
	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
MF	0.86523	0.06183	0.94195	0.11521	0.90068	0.15532	0.96395	0.08342
SAM	0.58673	0.26238	0.98268	0.02645	0.98302	0.05615	0.99240	0.02663
CEM	0.85973	0.06604	0.94089	0.14025	0.87960	0.15790	0.96619	0.08563
E-CEM	0.75543	0.01389	0.79541	0.00544	0.79495	0.00233	0.85960	0.00451
SID	0.74931	0.06428	0.98384	0.00444	0.98862	0.00831	0.99242	0.00619
BLTSC	0.95077	0.02089	0.94155	0.00467	0.99114	0.00518	0.98199	0.00685
HTD-IRN	0.94569	0.12119	0.99138	0.05967	0.97947	0.16160	0.98937	0.02410
MLSN	0.85240	0.33002	0.97099	0.01437	0.94657	0.15737	0.98417	0.20798
HTD-WT	0.95857	**0.00176**	0.99136	0.00210	**0.99990**	0.00016	**0.99874**	0.00103
HTD-STF	**0.98289**	0.00580	**0.99928**	**0.00153**	0.99602	**0.00011**	0.99808	**0.00032**

TABLE II
AUC SCORES OF TEN DETECTION METHODS ON FIVE HYPERSPECTRAL DATASETS FOR HTD TASK

Methods	Urban-2		San Diego		HYDICE		EI Segundo		Pavia	
	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
MF	0.72550	0.06412	0.98013	0.17877	0.98716	0.13766	0.98628	0.31688	0.90793	0.29735
SAM	0.89443	0.02339	0.99518	0.04154	0.99103	0.03740	0.88378	0.08904	0.82142	0.05359
CEM	0.72021	0.06692	0.98154	0.17480	0.99062	0.18294	0.98680	0.31139	0.91942	0.20511
E-CEM	0.70158	0.01165	0.76768	0.00573	0.89290	0.00093	0.88279	0.03329	0.80282	0.00317
SID	0.86479	0.00454	0.99540	0.00606	0.98037	0.00500	0.82547	0.03528	0.82386	0.01385
BLTSC	0.76953	0.01319	0.98728	0.00662	0.99430	0.00815	0.98537	0.11447	0.90760	0.03598
HTD-IRN	0.98588	0.12680	0.98423	0.04598	0.99461	0.18690	0.98079	0.11770	0.93790	0.07010
MLSN	0.91170	0.13999	0.99109	0.05910	0.99981	0.15471	0.93090	0.08649	0.83743	0.18264
HTD-WT	0.89833	0.00321	**0.99803**	**0.00286**	0.99975	0.00083	0.99513	0.00642	**0.95969**	0.00922
HTD-STF	**0.99275**	**0.00017**	0.99598	0.00341	**0.99985**	**0.00006**	**0.99652**	**0.00312**	0.95516	**0.00013**

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT STAGES ON SIX HYPERSPECTRAL DATASETS IN THE AUC SCORES

Results	Airport-2		Beach		Urban-1		San Diego		HYDICE		Pavia	
	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
C	0.94064	0.14022	0.87951	0.15789	0.96660	0.08562	0.98159	0.17477	0.99052	0.18291	0.91942	0.20511
R	0.99909	0.00937	0.99598	0.00046	0.99782	0.00143	0.99596	0.00955	0.99984	0.00020	0.95439	0.00049
Detc	**0.99928**	**0.00153**	**0.99602**	**0.00011**	**0.99808**	**0.00032**	**0.99598**	**0.00341**	**0.99985**	**0.00006**	**0.95516**	**0.00013**

more on the sparsely distributed target pixels and effectively suppress background interference, realizing clearer separation between target and background pixels. This is attributed to the fact that SSRE was fully pretrained on the large-scale dataset HyperMix-10K in the pretraining stage, which allows it to extract discriminative features of HSIs.

To quantitatively evaluate the different methods, the AUC values of the different methods on eight HSI datasets are calculated. The results are shown in Tables I and II, where the highest values are highlighted in bold, and the suboptimal values are underlined. It is evident that the proposed HTD-STF outperforms the other methods. The detailed analysis is given as follows. As for the traditional methods, the CEM and MF show similar detection performance, and their AUC_1 and AUC_2 values are always close. However, their performance is generally weaker than the SAM on most testing datasets. In contrast, the E-CEM and SID show lower AUC_2 values than other traditional methods on most datasets, indicating a better ability for BS. As for DL-based methods, these methods can achieve higher AUC_1 values on most datasets than traditional methods. However, the proposed HTD-STF shows stronger target enhancement ability than other competing methods, which attains the highest AUC_1 values on most of the datasets. And the AUC_2 values obtained by the proposed HTD-STF are lower than other competing methods in most cases, especially on the HYDICE dataset, which reveals the strong BS ability of the proposed HTD-STF. Note that AUC improvements in the range of 0.01–0.001 are highly significant in HTD because a higher AUC_1 and a lower AUC_2 indicate better detection performance. Therefore, the proposed HTD-STF has advantages in achieving higher AUC_1 values and lower AUC_2 values.

Table III shows the experimental results on six HSI datasets, including the coarse results **C**, the prediction results **R**, and the final detection results **Detc**. As illustrated in AUC_1, both the prediction results **R** and the final detection results **Detc** achieve higher AUC_1 values compared to the coarse results **C**. Meanwhile, as illustrated in AUC_2, the pixel-level fusion based on the coarse result significantly strengthens the background suppressibility of the whole framework. In addition, as observed in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

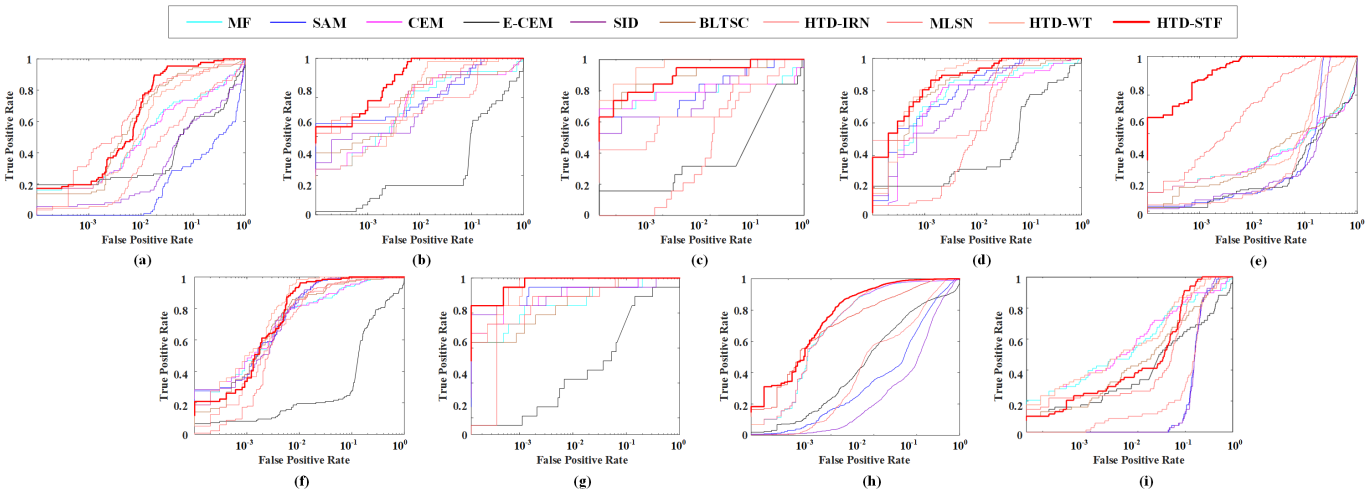


Fig. 8. ROC curves of (FPR and TPR) of detection results from ten detection methods on nine hyperspectral datasets for HTD task. (a) Airport-1. (b) Airport-2. (c) Beach. (d) Urban-1. (e) Urban-2. (f) San Diego. (g) HYDICE. (h) EI Segundo. (i) Pavia.

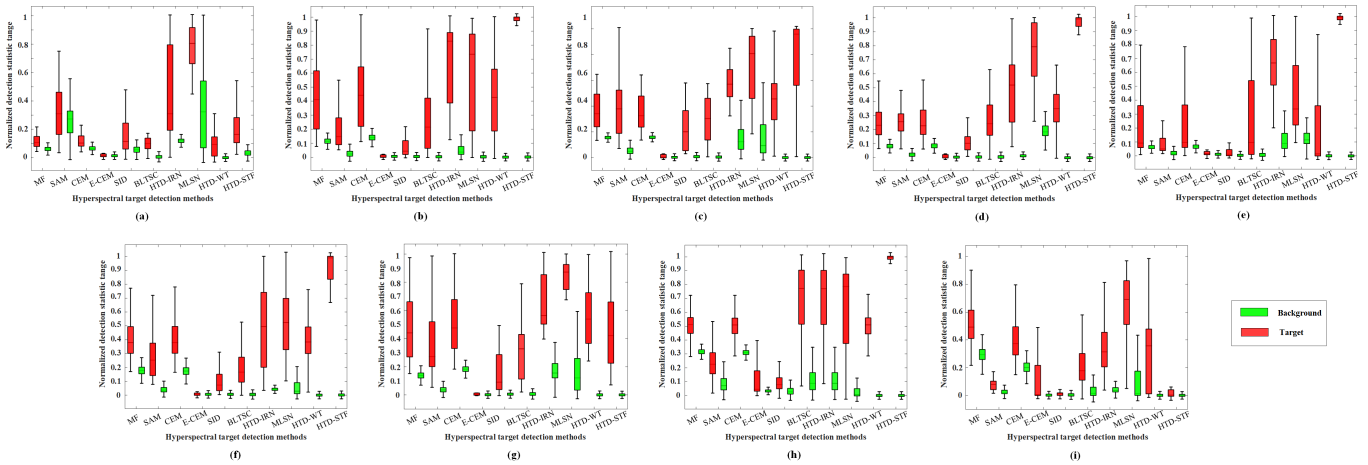


Fig. 9. Separability maps of detection results from ten detection methods on nine hyperspectral datasets for HTD task. (a) Airport-1. (b) Airport-2. (c) Beach. (d) Urban-1. (e) Urban-2. (f) San Diego. (g) HYDICE. (h) EI Segundo. (i) Pavia.

Table III, the original CEM detector is a representative method which can detect potential targets from spectral information well. However, it does not make full use of the spatial correlation among pixels in HSIs, resulting in lower AUC_1 and AUC_2 values compared to the prediction results and the final detection results. However, the proposed method effectively extracts spectral–spatial features from the pretrained encoder and realizes better target-background separation. Therefore, the AUC_1 and AUC_2 values of the prediction results outperform those of the coarse results. And the signature-constrained inference fusion can effectively suppress the background by combining the discriminative feature from the spectral and spatial domains. Therefore, it can be concluded that the contribution of the coarse result is significant, which helps our method makes full use of the spectral–spatial information and produces detection results with complementary information to coarse results.

Fig. 8 presents the ROC curves of (FPR and TPR) of detection results from different methods on eight datasets. The closer to the top-left corner of these curves, the better the detection performance is. It can be observed that the ROC curves of the HTD-STF are closer to the top-left corner than those of other competing methods, except for the Pavia dataset. On the Urban-2 and the HYDICE datasets, the ROC curves of the HTD-STF are always above other competing methods, indicating better detection accuracy. On other datasets, the HTD-STF initially shows lower detection probabilities than other methods. When the FPR exceeds the values of 0.01 and 0.001, the red curve representing the HTD-STF obviously surpasses other curves. Therefore, the proposed HTD-STF proves to be more reliable than other methods.

Fig. 9 shows the target-background separability plot of detection results from different methods, which can compare their ability to separate targets from backgrounds. The greater the gap between the red box (representing the target) and the green box (representing the background), the better the target-background separability of the detector. The HTD-STF outperforms other methods in separating background from targets, particularly on the Airport-2, Urban-2, and EI Segundo datasets, where the distance between the two boxes is larger.

For DL-based methods, the time of model training and testing is meaningful. We record the fine-tuning time of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
TRAINING TIME (IN SECONDS) OF DL-BASED METHODS FOR NINE HYPERSPECTRAL DATASETS

Methods	Airport-1	Airport-2	Beach	Urban-1	Urban-2	San Diego	HYDICE	EI Segundo	Pavia
BLTSC	52.4376	50.3124	74.1022	49.9914	52.5441	49.5818	45.7367	588.3514	73.2164
HTD-IRN	912.2541	882.4795	184.9150	761.4243	751.8085	148.9244	757.8935	568.7841	182.8467
MLSN	607.3107	548.61	527.36	609.7014	563.19	522.50	571.73	607.7634	598.2912
HTD-WT	498.7703	490.1735	1098.4372	495.2990	507.6564	507.9497	396.9928	3813.3050	1086.7442
HTD-STF	**40.03128**	**35.0877**	**70.3669**	**44.2349**	**51.3513**	**35.5488**	**38.9728**	**539.2696**	**69.4763**

TABLE V
TESTING TIME (IN SECONDS) OF DL-BASED METHODS FOR NINE HYPERSPECTRAL DATASETS

Methods	Airport-1	Airport-2	Beach	Urban-1	Urban-2	San Diego	HYDICE	EI Segundo	Pavia
BLTSC	2.0955	1.9125	2.2393	2.1337	2.0585	2.1245	1.9857	20.1941	2.1702
HTD-IRN	**0.6091**	**0.5410**	**0.4174**	**0.3481**	**0.3624**	**0.4762**	**0.3852**	**12.6262**	**0.3866**
MLSN	22.1251	21.7360	44.2006	1.9654	1.9806	1.9760	2.1117	18.2037	46.8390
HTD-WT	40.4734	40.5583	186.1256	39.3428	39.3681	36.6550	26.0187	2137.1302	187.9775
HTD-STF	1.8972	1.9028	1.7966	1.9654	1.9806	1.9760	2.1117	18.2037	1.7454

TABLE VI
AUC VALUES FOR COMPREHENSIVE ABLATION STUDY IN THE HTD-STF

Components	Airport-1		Urban-2		San Diego		HYDICE	
	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2	AUC_1	AUC_2
our − pre-training	0.54597	0.00050	0.69996	0.05150	0.83150	0.13975	0.88104	0.00752
our − SPL	0.96433	0.04479	0.99909	0.01122	0.99153	0.01044	0.97304	0.00842
our − SCFLF	0.92139	0.00062	0.93772	0.03216	0.92967	0.02063	0.93287	0.00948
our − fusion	0.98275	0.03507	0.99949	0.00143	0.99596	0.00955	0.99984	0.00020
our	**0.98289**	**0.00580**	**0.99275**	**0.00017**	**0.99598**	**0.00341**	**0.99985**	**0.00006**

HTD-STF as training time. Tables IV and V summarize the training and testing time for different DL-based methods on eight datasets, where the highest values are highlighted in bold, and the suboptimal values are underlined. As Table IV shows, the proposed HTD-STF is much faster than other methods during the fine-tuning stage. On the one hand, the HTD-STF inherits the powerful feature extraction capabilities of the encoder fully pretrained on the HyperMix-10K dataset. On the other hand, the pretrained encoder allows HTD-STF to perform well with minimal parameter adjustments during the fine-tuning. Although the HTD-WT also utilizes the pretrained encoder, it is only pretrained on training samples generated by testing HSI, resulting in a longer training time. Since MLSN uses labeled source data for meta-learning, it is still time-consuming. As shown in Table V, although the HTD-STF is not the fastest method in the testing stage, it is still faster than the BLTSC, MLSN, and HTD-WT.

D. Ablation Experiment

To fully explore the effectiveness of the proposed HTD-STF, the detection performance of different components of HTD-STF is evaluated on four HSI datasets: Airport-1, Urban-2, San Diego, and HYDICE. To mitigate the effects of random factors, five consecutive executions of the code were performed, and the average AUC values were obtained. Table VI shows the AUC scores of the ablation study of the HTD-STF, where the highest values are highlighted in bold, and the suboptimal values are underlined.

1) Pretraining Stage: Compared with other DL-based methods, the HTD-STF can achieve excellent detection performance on HTD tasks with a single pretraining. To verify the effectiveness of the pretraining stage, the corresponding ablation experiments were performed on four datasets. For convenience, the HTD-STF is denoted as our before ablation and our−pretraining after ablation. Table VI shows the AUC values of the ablation study for the pretraining stage in the HTD-STF. It can be found that without pretraining, the AUC_1 values drop significantly on all datasets, which indicates that the pretraining stage is significant for the whole framework to achieve the outstanding detection performance.

2) SPL Term in Object Function: To verify the effectiveness of the SPL function in the pretraining stage, the corresponding ablation experiments were performed on four datasets. For convenience, the HTD-STF is denoted as our before ablation and our−SPL after ablation. Table VI shows the AUC values of the ablation study for the SPL function. It can be found that without the SPL function, the performance on four datasets will decrease, which indicates that the SPL function helps the HTD-STF achieve robust detection performance on four datasets. During the pretraining, the SPL function ensures the smoothness of the reconstructed spectral curves by penalizing abrupt changes between adjacent bands, which is beneficial for achieving excellent spectral–spatial reconstruction.

3) SCFLF Module: To verify the effectiveness of the proposed SCFLF module, the corresponding ablation experiments were performed on four datasets. For convenience, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

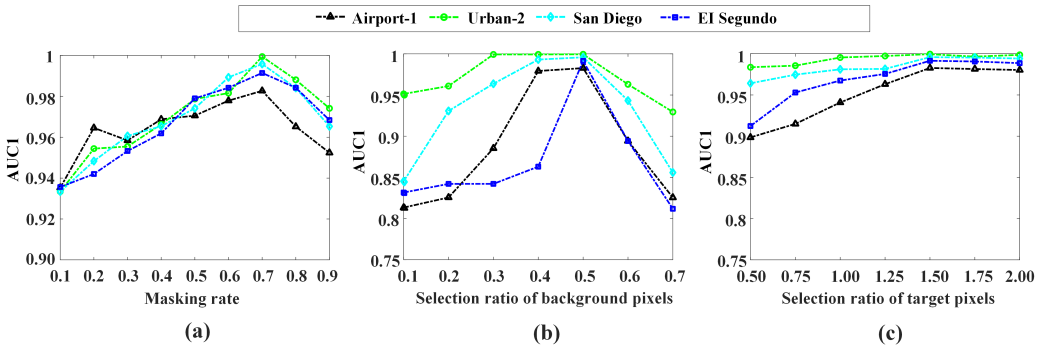


Fig. 10. Parameter sensitivity analysis of (a) spatial–spectral masking rate, (b) selection ratio of pseudo-background pixels, and (c) selection ratio of pseudo-target pixels.

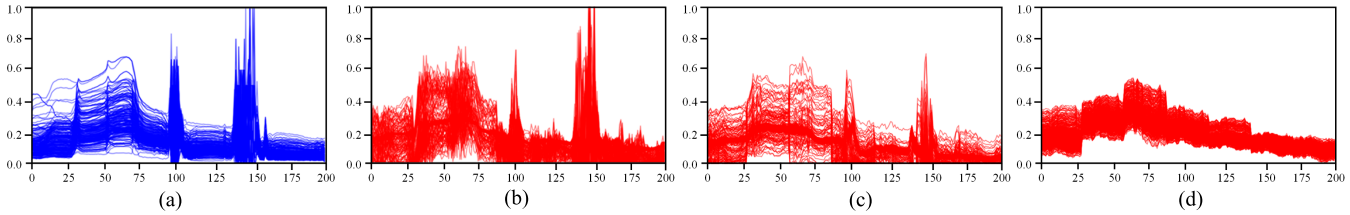


Fig. 11. Spectra of the selected region under different conditions. (a) Original spectra of the selected region, (b) $\lambda = 100$, (c) $\lambda = 10$, and (d) $\lambda = 0$.

HTD-STF is denoted as our before ablation and our−SCFLF after ablation. Table VI shows the AUC values of the ablation study for the SCFLF module during the fine-tuning. It can be found that without the SCFLF module, the performance on four datasets will decrease, which indicates that the SCFLF module helps the HTD-STF achieve high detection accuracy. This is because the SCFLF module uses the prior target signature to realize feature-level fusion with the encoded feature vectors from SSRE, which helps the HTD-STF achieve high detection accuracy.

4) Signature-Constrained Inference Fusion: To verify the effectiveness of the proposed signature-constrained inference fusion, the corresponding ablation experiments were performed on four datasets. For convenience, the HTD-STF is denoted as our before ablation and our−fusion after ablation. Table VI shows the AUC values of the ablation study for the signature-constrained inference fusion. It can be found that without fusion, the performance on the four datasets will decrease, which indicates that the fusion effectively suppresses the background by combining the discriminative feature from the spectral and spatial domains.

E. Parameter Sensitivity

This section analyzes the sensitivity of two important parameters of the proposed HTD-STF, i.e., the spectral–spatial masking rate adopted in the pretraining stage and the selection ratio of pseudo-background pixels adopted for CSMS in the fine-tuning stage. Specifically, comprehensive experiments were performed on four HSI datasets: Airport-1, Urban-2, San Diego, and HYDICE. To mitigate the effects of random factors, five consecutive executions of the code were performed, and the average AUC values were obtained.

First, the spectral–spatial masking rate in the pretraining stage was set to 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, and 0.9. Fig. 10(a) shows the AUC values for parameter analysis of masking rate on four datasets. It can be observed that the detection accuracy increases with the increase of the masking rate, and reaches a maximum when the masking rate is 0.70, then decreases with the increase of the masking rate. A smaller masking rate makes it easier to complete the spectral–spatial reconstruction since it contains more encoded feature vectors from SSRE. However, this will lead to lower detection accuracy because the feature extraction ability of SSRE has not been fully explored. Besides, when the masking rate exceeds a certain limit, the detection accuracy will be lower because the SSRE copes with fewer encoded features. Hence, we set the best masking rate as 0.70 in this work.

Second, the selection ratio of pseudo-background pixels in CSMS during the fine-tuning was set to 0.1, 0.2, 0.3, 0.4, and 0.5. The AUC values for parameter analysis of the selection ratio of pseudo-background pixels on four datasets are shown in Fig. 9(b). On four datasets, the average value of AUC_1 is above 0.8. The AUC curve presents an overall upward trend, reaching the peak AUC_1 value when the selection ratio is 0.5. It indicates that the higher the proportion of background pixels, the better BS effect the HTD-STF can achieve. However, when the background pixels continue to increase, the AUC_1 value begins to decrease, indicating that the model is disturbed by a large number of background pixels during the learning process. Hence, we set the best selection ratio of pseudo-background pixels as 0.5 in this work.

Third, the AUC values for parameter analysis of the selection ratio of pseudo-target pixels on four datasets are shown in Fig. 9(c). The selection ratio of target pixels in CSMS during the fine-tuning was set to 0.50, 0.75, 1.00, 1.25, 1.50, 1.75,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

and 2.00 in percentage. On four datasets, the average value of AUC_1 is above 0.9. The AUC curve presents an overall upward trend, reaching the peak AUC_1 value when the selection ratio is 1.5%. As the ratio continues to increase, the AUC_1 value decreases slightly, indicating that the model has gained a strong feature extraction ability. Hence, we set the best selection ratio of pseudo-target pixels as 1.5% in this work.

Fourth, to determine the best parameter λ for SPL, the λ is set as $100, 10$, and 0. The quality of reconstructed spectra in the selected region is considered as the criterion. As Fig. 11 shows, when $\lambda = 100$, the reconstructed spectral curves closely match the original curves in both local details and global trends, which indicates that our method achieves superior spectral curve reconstruction quality, Therefore, the high λ amplifies the role of SPL in penalizing spectral variance discrepancies across the entire tensor group, which is critical for high-quality spectral–spatial reconstruction. Hence, we set the best parameter λ for SPL as 100 in this work.

IV. Conclusion

A novel signature-constrained STF for HTD based on GSSL, the HTD-STF, is proposed in this article. To liberate the HTD model from dependence on the quantity and quality of training samples, a GSSL-based pretraining stage is conducted at first on the newly constructed large-scale HSI dataset, the HyperMix-10K. HTD-STF can learn the universal feature representation of diverse scenes with unlabeled HSIs. The pretrained encoder can be adapted to different HSIs via low-cost fine-tuning. The second stage utilizes a novel signature-constrained fine-tuning strategy for preprocessing, training, and inference. HTD-STF makes full use of the prior information and obtain the feature-level and pixel-level guidance to highlight the target information, which improves detection performance. Extensive experimental results verify the superiority of the proposed HTD-STF in terms of accuracy and generalization. The introduction of the HyperMix-10K dataset provides a valuable resource for further research in GSSL-based HTD. On this basis, future work will focus on model light-weighting and framework optimization.

References

[1] L. Sun, Z. Ma, and Y. Zhang, "ABLAL: Adaptive background latent space adversarial learning algorithm for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 411–427, 2024.

[2] T. Guo, F. Luo, J. Guo, Y. Duan, X. Huang, and G. Shi, "Hyperspectral target detection with target prior augmentation and background," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1765–1780, 2024.

[3] Y. Shi, X. Li, and S. Chen, "Target-driven iterative autoencoder for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505518.

[4] X. Zhao, Z. Hou, X. Wu, W. Li, P. Ma, and R. Tao, "Hyperspectral target detection based on transform domain adaptive constrained energy minimization," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102461.

[5] H. Huang et al., "Underwater hyperspectral imaging for in situ underwater microplastic detection," *Sci. Total Environ.*, vol. 776, Jul. 2021, Art. no. 145960.

[6] X. Sun, H. Zhang, F. Xu, Y. Zhu, and X. Fu, "Constrained-target band selection with subspace partition for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9147–9161, 2021.

[7] C. Li, D. Zhu, C. Wu, B. Du, and L. Zhang, "Global overcomplete dictionary-based sparse and nonnegative collaborative," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5513214.

[8] S. Zhang, Y. Mo, X. Kang, and S. Li, "Exposure fusion-based shadow-insensitive hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502511.

[9] C.-I. Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505223.

[10] F. Kruse et al., "The spectral image processing system interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, no. 2, pp. 145–163, 1993.

[11] C.-I. Chang, "An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1927–1932, Aug. 2000.

[12] Q. Du, H. Ren, and C.-I. Chang, "A comparative study for orthogonal subspace projection and constrained energy minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1525–1529, Jun. 2003.

[13] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems: An overview of practical algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.

[14] D. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.

[15] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, May 1992.

[16] C.-I. Chang and D. Heinz, "Constrained subpixel target detection for remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1144–1159, May 2000.

[17] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, Jun. 2019.

[18] L. Capobianco, A. Garzelli, and G. Camps-Valls, "Target detection with semisupervised kernel orthogonal subspace projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3822–3833, Nov. 2009.

[19] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[20] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[21] A. Creswell et al., "Generative adversarial networks: An overview," *IEEE Signal Process. Mag.*, vol. 35, no. 1, pp. 53–65, Jan. 2018.

[22] A. Vaswani, "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 1–11.

[23] Y. Li, Y. Shi, K. Wang, B. Xi, J. Li, and P. Gamba, "Target detection with unconstrained linear mixture model and hierarchical denoising autoencoder in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 31, pp. 1418–1432, 2022.

[24] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020.

[25] Y. Wang, X. Chen, F. Wang, M. Song, and C. Yu, "Meta-learning based hyperspectral target detection using Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527913.

[26] W. Xie, X. Zhang, Y. Li, K. Wang, and Q. Du, "Background learning based on target suppression constraint for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5887–5897, 2020.

[27] H. Qin, W. Xie, Y. Li, and Q. Du, "HTD-TS3: Weakly supervised hyperspectral target detection based on transformer via spectral–spatial similarity," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16816–16830, Nov. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

[28] Y. Li, H. Qin, and W. Xie, "HTDFormer: Hyperspectral target detection based on transformer with distributed learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524715.

[29] W. Xie, J. Yang, J. Lei, Y. Li, Q. Du, and G. He, "SRUN: Spectral regularized unsupervised networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1463–1474, Feb. 2020.

[30] Y. Shi, J. Li, Y. Zheng, B. Xi, and Y. Li, "Hyperspectral target detection with RoI feature transformation and multiscale spectral attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5071–5084, Jun. 2021.

[31] D. Shen, X. Ma, W. Kong, J. Liu, J. Wang, and H. Wang, "Hyperspectral target detection based on interpretable representation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519416.

[32] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Jan. 2023.

[33] J. M. Haut, M. E. Paoletti, S. Moreno-Alvarez, J. Plaza, J.-A. Rico-Gallego, and A. Plaza, "Distributed deep learning for remote sensing data interpretation," *Proc. IEEE*, vol. 109, no. 8, pp. 1320–1349, Aug. 2021.

[34] X. Chen, Y. Zhang, Y. Dong, and B. Du, "Spatial–spectral contrastive self-supervised learning with dual path networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515612.

[35] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510515.

[36] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 197–211.

[37] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.

[38] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[39] X. Chen, Y. Zhang, Y. Dong, and B. Du, "Generative self-supervised learning with spectral–spatial masking for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522713.

[40] H. Qin, S. Wang, Y. Li, W. Xie, K. Jiang, and K. Cao, "Hyperspectral target detection based on generative self-supervised learning with wavelet transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510815.

[41] Y. Niu and B. Wang, "Extracting target spectrum for hyperspectral target detection: An adaptive weighted learning method using a self-completed background dictionary," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1604–1617, Mar. 2017.

[42] X. Zhao, J. Huang, Y. Gao, and Q. Wang, "Hyperspectral target detection based on prior spectral perception and local graph fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13936–13948, 2024.

[43] H. Qin, W. Xie, Y. Li, K. Jiang, J. Lei, and Q. Du, "Weakly supervised adversarial learning via latent space for hyperspectral target detection," *Pattern Recognit.*, vol. 135, Mar. 2023, Art. no. 109125.

[44] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2020, pp. 1–22.

[45] Z.-H. Zhou, "A brief introduction to weakly supervised learning," *Nat. Sci. Rev.*, vol. 5, no. 1, pp. 44–53, Jan. 2018.

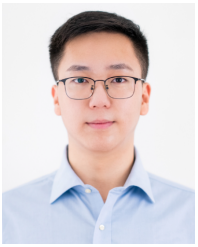
[46] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[47] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.

[48] B. Kunkel, F. Blechinger, R. Lutz, R. Doerffer, H. van der Piepen, and M. Schröder, "ROSIS (reflective optics system imaging spectrometer)—A candidate instrument for polar platform missions," *Proc. SPIE*, vol. 868, pp. 134–141, Apr. 1988.

[49] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognit. Lett.*, vol. 27, no. 8, pp. 861–874, Jun. 2006.

[50] J. Huang and C. X. Ling, "Using AUC and accuracy in evaluating learning algorithms," *IEEE Trans. Knowl. Data Eng.*, vol. 17, no. 3, pp. 299–310, Mar. 2005.



Haonan Qin (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2023.

He is an Associate Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. He has published more than ten articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Pattern Recognition*, *Remote Sensing*, Conference on IEEE International Geoscience and Remote Sensing Symposium (IGARSS), and IEEE International Conference on Tools with Artificial Intelligence (ICTAI). His research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.

Shuai Wang (Graduate Student Member, IEEE) received the B.E. degree in telecommunication engineering from Xidian University, Xi'an, China, in 2023, where he is currently pursuing the Ph.D. degree in communication and information systems.

He works as a Doctoral Student with the State Key Laboratory of Integrated Services Networks, Xidian University. His research interests include machine learning and hyperspectral image processing.

Yunsong Li (Member, IEEE) received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2002.

He is a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University, where he is also the Director of the Image Processing and Image Communication (IPIC) Laboratory, State Key Laboratory of Integrated Services Networks. He has published articles in refereed journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, *Neural Networks*, *Pattern Recognition*, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING . His research interests include image and video processing and high-performance computing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.

Weiying Xie (Senior Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2017.

She is a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 50 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, Conference on IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR), and Association for the Advancement of Artificial Intelligence (AAAI). Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.

Kai Jiang received the B.E. degree in information engineering and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2019 and 2024, respectively.

His research interests include image processing and deep learning.

Kailang Cao received the B.E. degree from the School of Electronic and Information Engineering, Tiangong University, Tianjin, China, in 2017, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2023.

He currently holds a post-doctoral position at the School of Telecommunication, State Key Laboratory of Integrated Services Networks, Xidian University. His research interests include remote sensing survey and mapping, 3-D reconstruction, and deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:57:38 UTC from IEEE Xplore. Restrictions apply.